

Arion Stern

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Education

University of Florida – Herbert Wertheim College of Engineering

B.S. Computer Engineering · Expected Graduation: May 2028 · GPA: 4.00 · Credits: 87

Scholarships & Honors: University of Florida Distinguished Scholar Waiver, UF Dean's List (Fall 2024, Spring 2025, Spring 2026), Eta Kappa Nu (HKN) IEEE Honor Society (Spring 2026)

Relevant Courses: Microprocessor Applications, Digital Logic & Computer Systems, Digital Design, Signals & Systems, Applied Machine Learning, Data Structures & Algorithms, Advanced Programming Fundamentals, Discrete Structures, Calculus II–III, Physics II, Diff Eq

Pascack Hills High School – Montvale, NJ (2020-2024) 4.75/4.00 (W/UW)

Honors: AP Scholar with Distinction, High Honor Roll (all years), National Honor Society, Vice President of Senior Class, Dr. Michael Weinthal Memorial Scholarship Recipient

Technical Skills

- **Hardware:** FPGA (DE10-Lite), ATxmega128A1U Microcontroller, Arduino, Breadboarding, Oscilloscope, Soldering
- **Tools:** Quartus Prime, QuestaSim, ModelSim, Altium Designer, Microchip Studio, MATLAB, Git, WaveForms
- **Languages:** Assembly (G-CPU & AVR), VHDL, SystemVerilog, Python, C++, C, Java, MATLAB
- **Concepts:** Embedded Systems, RTL Design, CPU Architecture, FSMs, Datapath/Control Logic, SVA/CRV Verification, Machine Learning, Computer Vision

Projects

[Computer Vision-Driven Color Detection System \(Python, OpenCV, Arduino\)](#)

- Built a real-time color-detection pipeline using OpenCV (HSV space, noise-averaged ROI sampling) and streamed color data to an Arduino over serial. Designed an embedded UI with RGB LED modes (vision, rainbow, gator-theme), LCD display pages (RGB/HSV/history), and button-controlled interaction.

[Pong — FPGA VGA Game \(VHDL, DE10-Lite\)](#)

- Implemented a complete 2-player Pong game entirely in synthesized hardware (no CPU, no software) as extra credit for EEL4712C.
- VGA output at 640x480 @ 60 Hz with ball physics, paddles, dashed center line, scoreboard, and bitmap-rendered screens; FSM with next-position lookahead collision detection.

[8-Bit Simple CPU \(FPGA, Quartus/ModelSim\)](#)

- Built a CPU with 8 custom instructions, microprogrammed control, and memory-mapped I/O.
- Developed datapath (ALU, registers, PC), wrote assembly for branching and indirect memory access, verified via simulation and FPGA.

[Microprocessor Applications \(ATxmega128A1U, C & AVR Assembly, Microchip Studio\)](#)

- Programmed embedded systems integrating digital I/O, timer interrupts, and PWM to control RGB LED intensity and implement precise timing and event-driven logic on the µPAD and Switch & LED Backpack.
- Interfaced with an LSM6DSL accelerometer and gyroscope via SPI to stream real-time 3-axis data through USART, sampled light-sensor signals using the ADC for visualization, and generated DAC-based sine waveforms with DMA for an audio synthesizer.

[Elevation Sort Visualizer \(Python, Pygame\)](#)

- Created an interactive tool to animate sorting algorithms on NOAA elevation data.
- Implemented six algorithms (Bubble, Insertion, Selection, Merge, Quick, Heap) with real-time metrics, tooltips, and user controls.

[Drone Payload Deployment System \(Arduino, Hardware Build\)](#)

- Designed a lightweight payload release mechanism with 3D-printed parts and Arduino-controlled servos.
- Integrated embedded logic for altitude-triggered deployment.

Leadership & Organizations

- **Swamp Launch** – IREC Avionics; designed flight sensor PCB in Altium (schematic, layout, I²C/UART integration). (Fall 2025)
- **Dream Team Engineering — OR Table Project** – Developing a Raspberry Pi camera system to track surgical instruments on an OR table; integrated GPIO button/LED sync with video recording for a UF surgical waste-reduction initiative. (2025–Present)
- **TEA Lab — Autonomous Systems Research** – Implemented DVBF and SINDYc trajectory prediction baselines for a PIWM autonomous driving research project (DonkeyCar); built VAE encoding, rollout evaluation, and visualization pipeline. (2025)
- **IEEE** - Member (2024-present)
- **Phi Kappa Tau Fraternity** – Member (2024–Present)
- **Volunteer:** Project Paterson, Montvale Baseball Clinic, Tri-Boro Food Pantry, Talbot Elementary STEM Outreach

Experience

Lifeguard Instructor – Flagship First, NJ · Summer 2023

- Taught rescue techniques, CPR, and supervised training simulations.

Camp Counselor – Park Ridge Recreation · Summer 2022

- Supervised and led activities for children, ensuring safety and engagement.

Video Auditor – Laundry Capital / Mr. Jeff LLC · 2020–2021

- Conducted loss prevention audits and reported findings in organized spreadsheets.